README

This folder contains the code to replicate all results in the paper and the cascad reproducibility certificate. Additional code and data are available at https://github.com/eguidotti/bidask

Data Availability

The paper is based on the following data sources:

1. CRSP U.S. Stock Database (flat files for R)
2. NYSE Monthly and Daily TAQ (via WRDS)
3. Binance Historical Market Data (free access)

The data cannot be redistributed.

Dataset List

The folder `input/` includes pseudo datasets. The pseudo datasets have the same structure as the original datasets used for the paper but they have a reduced number of entries and are filled with random values.

File	Location	Provided	Note
`stocks.csv`	`input/`	TRUE	Pseudo data
`cryptos.csv`	`input/`	TRUE	Pseudo data

To overwrite the pseudo data with the actual data, follow the instructions:

1. in the folder `data/source/crsp/202112/` to generate the files `frachl.csv.gz` and `spread.csv.gz`
2. in the folder `data/source/crsptaq/202112/` to generate the file `ctm_mth.csv.gz`
3. in the folder `data/source/binance/202302/` to generate the file `bmd_mth.csv.gz`

Then, run the script `stocks.R` to generate the actual data file `input/stocks.csv`:

```
Rscript stocks.R
```

and the script `cryptos.R` to generate the actual data file `input/cryptos.csv`.

```
Rscript cryptos.R
```

Computational Requirements

All results were generated using R 4.2.3 on a MacBook Air (M1, 2020).

Instructions to Replicators

Install all dependencies with renv:

```
renv::restore()
```

Run the main script to generate all results in the paper:

```
Rscript main.R
```

The expected runtime is 10-20 minutes. The script generates an `output/` folder with all figures and tables.

List of Tables and Programs

Figure/Table	Program	Line number	Output file	Dataset used
Figure 1	`main.R`	194	`fig_1.pdf`	`stocks.csv`
Figure 2	`main.R`	162	`fig_2.pdf`	
Figure 3	`main.R`	176	`fig_3.pdf`	
Figure 4	`main.R`	228	`fig_4.pdf`	`stocks.csv`
Figure 5	`main.R`	236	`fig_5.pdf`	`stocks.csv`
Figure 6	`main.R`	307	`fig_6.pdf`	`stocks.csv`
Figure 7	`main.R`	314	`fig_7.pdf`	`stocks.csv`
Figure 8	`main.R`	321	`fig_8.pdf`	`cryptos.csv`
Table 2 / Panel A	`main.R`	134	`tab_2A.tex`	
Table 2 / Panel B	`main.R`	148	`tab_2B.tex`	
Table 3	`main.R`	200	`tab_3.tex`	`stocks.csv`
Table 4 / Panels A-E	`main.R`	244	`tab_4(A-E).tex`	`stocks.csv`
Section 6.2 - Text	`main.R`	207	`sec_6.2.tex`	`stocks.csv`
Internet Appendix				
Figure I.2	`main.R`	221	`fig_I.2.pdf`	`stocks.csv`
Figure I.3	`main.R`	328	`fig_I.3.pdf`	`stocks.csv`
Table I.2	`main.R`	214	`tab_I.2.tex`	`stocks.csv`
Table I.4 / Panels A-E	`main.R`	253	`tab_I.4(A-E).tex`	`stocks.csv`
Table I.5 / Panels A-E	`main.R`	262	`tab_I.5(A-E).tex`	`stocks.csv`
Table I.6 / Panels A-E	`main.R`	271	`tab_I.6(A-E).tex`	`stocks.csv`
Table I.7 / Panels A-E	`main.R`	280	`tab_I.7(A-E).tex`	`stocks.csv`
Table I.8 / Panels A-E	`main.R`	298	`tab_I.8(A-E).tex`	`stocks.csv`
Table I.9 / Panels A-E	`main.R`	289	`tab_I.9(A-E).tex`	`stocks.csv`